

APIs and conventional data pipelines to access enterprise systems, which creates bottlenecks that limit their autonomous capabilities.

The polling problem

Without real-time event notifications, agents resort to polling: repeatedly querying APIs to detect changes. A fraud detection agent polling a transaction feed every few seconds is still operating on stale data. More fundamentally, polling locks agents into a reactive posture at the exact moment they need to be proactive.

Synchronous brittleness

When an agent's workflow depends on sequential API calls, a single slow or

unavailable service stalls everything downstream. This fragility is one reason why a Gartner survey found that, on average, only 48% of AI projects make it into production — and why the average journey from prototype to production takes eight months.

State fragmentation

Multiple agents calling different APIs at different moments will inevitably operate on different snapshots of reality. When agents make decisions based on inconsistent state, the results range from redundant actions to directly contradictory ones. In financial services, healthcare, and supply chain environments, this is not a theoretical concern.

APIs tell agents what exists. Events tell them when to act.

Events: the coordination layer agents need

Event-driven architecture (EDA) is not a new idea. Enterprises have used it for decades in domains where real-time responsiveness is non-negotiable: financial trading, logistics, telecommunications. What is new is its relevance to AI — specifically, its role as the coordination layer that autonomous agents need but API-only architectures cannot provide.

The core principle is straightforward. Rather than agents polling systems for updates, systems emit events when something meaningful

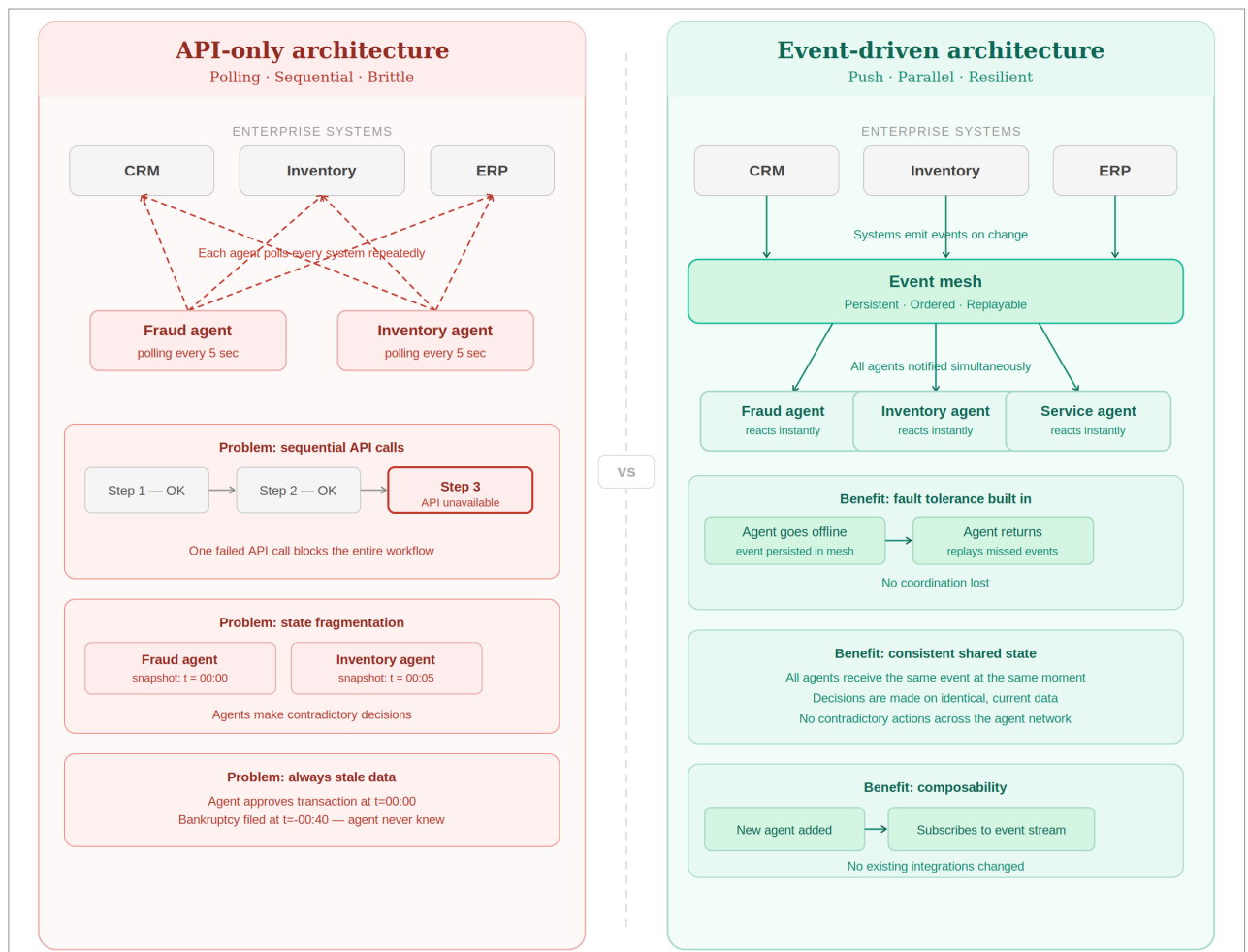


Figure 2: The four components of a production-grade agentic AI architecture